# Pharmaceutical Calculations Booklet

## Midterm 2 · Lecture 8

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Each lecture is split into topics. Every topic has **Lecture Problems**, then **Textbook Problems**. Answers are at the end of each lecture.

# 8 Intravenous Therapy

**In this lecture**


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## 8.1 Giving Sets & Drip Rate (drops/min)

### Lecture Problems

**Slide 4** Giving sets — how many drops are in 1 mL?

**a.** Standard giving set (SGS)

**b.** Microdrop giving set (burette)

**Slide 5** One litre of 5% dextrose is to be given over 8 hours. What drip rate is required? (rate: 20 drops/mL)

Drip rate = ?

### Textbook Problems

**Example 11.1 · Page 195** One litre of 0.9% saline solution is to be given to a patient over a 6-hour period. If 20 drops = 1 mL, how many drops per minute should be administered?

Drops/min = ?

**Q2 · Page 209** 500 mL of 0.9% saline solution is to be given to a patient over a 6-hour period. If 20 drops = 1 mL, calculate how many drops per minute should be administered to the patient.

Drops/min = ?

**Example 11.3 · Page 197** 20 mL of Addiphos solution and 10 mL of multivitamin infusion are added to 500 mL of glucose 5% solution. The resultant solution is to be administered over 4 hours. The administration set is calibrated to 20 drops/mL. Calculate the number of drops per minute to be given if the patient is to receive all of the solution in the specified time.

Drops/min = ?

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**Q4 · Page 209** A 10-mL vial of multivitamin injection is to be added to 500 mL of glucose 5% solution. The resultant solution is to be administered to a patient over 3 hours. The administration set is calibrated at 20 drops/mL. Calculate the number of drops per minute to be given if the patient is to receive all the solution in the specified time.

Drops/min = ?

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## 8.2 Converting mL/hour, mL/min & drops/min

### Lecture Problems

**Slide 6** One litre of sodium chloride 0.9% has to be given at a rate of 120 mL/hour. What is the rate in drops/min? (20 drops/mL)

Rate = ? drops/min

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### Textbook Problems

**Q1 · Page 209** 1 L of glucose 5% solution is to be given to a patient over a 5-hour period. Calculate the rate of delivery in mL/minute.

Rate = ? mL/min

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**Q3 · Page 209** An IV solution is to be administered at a rate of 2 mL/minute. What volume will be required if the infusion is to be maintained for 3 hours?

Volume = ?

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## 8.3 Dose Rates (mcg/kg/min) & mL/hour

**Lecture Problems**

**Slide 7** You have an infusion of dopamine 800 mg in 500 mL. The dose required is 2 mcg/kg/min for a patient weighing 68 kg. (20 drops/mL)

**a.** What is the rate in mL/hour?

**b.** What is the rate in drops/min?

**Slide 8** An infusion pump containing 250 mg dobutamine in 50 mL is administered at a rate of 3.5 mL/hour. Convert to mcg/kg/min. (Patient weight: 70 kg)

Dose = ? mcg/kg/min

## 8.4 Volume, Drip Rate & Duration

**Lecture Problems**

**Slide 9** A prescription of one litre 5% dextrose to be administered over 8 hours at a drip rate of 40 drops/min. You wish to check the drip rate. How long will it take for the infusion to finish? (20 drops/mL)

Time = ?

## Textbook Problems

**Example 11.2 · Page 196** The recommended flow rate for drug Y is 2–3 mL/minute at a concentration of 5 mg/mL. Drug Y is supplied in 500-mg vials for dilution in glucose 5%. Using one vial, the flow rate is 3 mL/minute and the IV set is calibrated to 20 drops/mL.

**a.** Volume of glucose 5% solution to use?

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**b.** Drops per minute?

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**c.** How long will the infusion take?

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**Q5 · Page 209** Nizatidine injection (25 mg/mL) is supplied in 4-mL vials. The BNF states that for continuous infusion "dilute 300 mg in 150 mL and give at a rate of 10 mg/hour". If this procedure is adopted, calculate:

**a.** How many vials are required?

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**b.** The flow rate (mL/hour)?

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**c.** The length of time the infusion will take?

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## 8.5 Dose, Volume & Maximum Rate

**Lecture Problems**

**Slide 10** A 73 kg patient requires drug X by IV infusion. The dose is 400 mcg/kg. The final concentration of solution must not exceed 150 mcg/mL. The drug is available as a vial of 120 mg. The infusion needs to be delivered in 3.5 hours.

1. Calculate the dose.

2. Calculate the volume required by the patient.

3. What is the rate in mL/min?

4. What is the total volume of the solution prepared (whole vial)?

**Slide 11** The required dose of metoprolol is 75 mg at a rate not exceeding 8 mg/min. Metoprolol injection is available as 20 mg/mL.

a. Volume of metoprolol injection required?

b. Infusion rate (mL/min)?

c. Infusion duration?

## Textbook Problems

**Example 11.4 · Page 198** A 56-kg female patient requires amphotericin by IV infusion at a dose of 250 micrograms/kg. The concentration of the final solution must not be more than 100 micrograms/mL. A vial of amphotericin contains 50 mg. The solution has to be delivered in 2.5 hours.

**a.** Dose = ?

**b.** Volume of IV solution (maximum concentration)?

**c.** Rate in mL/minute?

**d.** Total volume if a 50-mg vial is used?

**Example 11.6 · Page 201** The required dose of furosemide by slow IV infusion is 50 mg at a rate not exceeding 4 mg/minute. Furosemide injection contains 10 mg in 1 mL. Calculate the volume of injection required and the infusion rate (mL/minute).

**a.** Volume of injection?

**b.** Time and infusion rate?

**Q8 · Page 210** The recommended dose of drug X is 75 mg by slow infusion at a rate not exceeding 0.3 g/hour. Drug X is available as an injection containing 15 mg in 1 mL. Calculate the volume of injection required and the infusion rate in mL/minute.

Volume and rate = ?

**Example 11.5 · Page 200** Phenytoin: 18 mg/kg, rate not exceeding 50 mg/minute. Patient: woman, 56 kg. Injection: 5-mL ampoules, 50 mg/mL. The prescriber would like an infusion volume of 100 mL and a dose rate of 25 mg/minute. Is the request possible and, if so, what is the flow rate per minute?

**a.** Dose and volume of injection?

**b.** Possible? Flow rate?

**Q6 · Page 209** A pharmacist prepares a solution of vancomycin containing 1 g in 200 mL. The rate of infusion should not exceed 10 mg/minute.

**a.** Maximum flow rate (mL/min)?

**b.** Flow rate if given over 2 hours?

**Q7 · Page 209** A baby weighing 3.4 kg requires phenytoin at a dose of 20 mg/kg. A 5-mL vial of phenytoin injection containing 50 mg/mL is available. A suitable volume of injection is diluted up to 50 mL with sodium chloride 0.9%.

**a.** Volume of phenytoin injection to use?

**b.** Maximum flow rate if the rate must not exceed 3 mg/kg/minute?

**c.** Minimum flow rate if completed in 1 hour?

## 8.6 mg/hour → mL/min & drops/min

**Lecture Problems**

**Slide 12** A drug is available as an infusion solution at a concentration of 0.25 mg/mL. The prescribed flow rate is 5 mg/hour. Calculate the flow rate in mL/min and drops/min (1 mL = 20 drops).

Flow rate = ?

**Textbook Problems**

**Example 11.7 · Page 203** A drug is provided as an infusion solution at a concentration of 0.15 mg/mL. The prescribed flow rate is 3 mg/hour. Calculate the flow rate in mL/minute and drops/minute if 1 mL = 20 drops.

Flow rate = ?

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**Q9 · Page 210** A drug is provided as an infusion solution at a concentration of 2 micrograms/mL. The prescribed flow rate is 0.05 mg/hour. Calculate the flow rate in mL/minute and drops/minute if 1 mL = 25 drops.

Flow rate = ?

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## 8.7 Syringe Drivers (mm/hour)

**Textbook Problems**

**Example 11.8 · Page 204** Apomorphine injection (10 mg/mL) is diluted with an equal volume of sodium chloride 0.9% and put in a 10-mL syringe that is 60 mm long. The patient requires 2.5 mg of apomorphine per hour. At what rate (mm/hour) should the syringe driver be set?

Rate = ? mm/hour

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**Example 11.9 · Page 205** A 100-mg dose of an analgesic has to be infused over 5 hours. The drug is in 1-mL vials each containing 50 mg. The syringe driver holds 20 mL and delivers 2 mL/hour. How many vials and what volume of diluent are needed to fill the 20-mL syringe?

Vials and diluent = ?

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**Q10 · Page 210** A syringe driver is designed to contain 5 mL of injection and has a length of 60 mm. Injection Y contains 4 mg/mL. If a patient requires 2.5 mg of drug Y per hour, at what rate setting in mm/hour should the syringe driver be fixed?

Rate = ? mm/hour

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**Q11 · Page 210** A syringe driver is set at 5 mm/hour. The length of the syringe driver is 80 mm and it holds 10 mL of injection solution. The injection contains 5 micrograms of drug X in 1 mL. What dose of drug X will the syringe driver deliver in 3 hours?

Dose = ?

# Answers · Lecture 8

## 8.1 Giving Sets & Drip Rate (drops/min)

**Slide 4**

**a. 20 drops/mL for crystalloids · 15 drops/mL for colloids** — Crystalloids: e.g. sodium chloride, dextrose.

Colloids: e.g. blood.

**b. 60 drops/mL**

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**Slide 5**

**42 drops/min** — Step 1: 1000 mL × 20 = 20 000 drops

Step 2: 8 h × 60 = 480 min

Step 3: 20 000 / 480 = 41.67 → round up → 42 drops/min

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**Example 11.1 · Page 195**

**56 drops/min** — 6 h = 360 min

1000 mL × 20 = 20 000 drops

20 000 / 360 = 55.5 → 56 drops/min

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**Q2 · Page 209**

**28 drops/min** — 500 × 20 = 10 000 drops

6 × 60 = 360 min

10 000 / 360 = 27.8 → 28 drops/min

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**Example 11.3 · Page 197**

**44 drops/min** — Total volume = 20 + 10 + 500 = 530 mL

530 × 20 = 10 600 drops

4 × 60 = 240 min

10 600 / 240 = 44.17 → 44 drops/min

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**Q4 · Page 209**

**≈ 57 drops/min (56.7)** — Total volume = 10 + 500 = 510 mL

510 × 20 = 10 200 drops

3 × 60 = 180 min

10 200 / 180 = 56.7 → 57 drops/min

## 8.2 Converting mL/hour, mL/min & drops/min

**Slide 6**

**40 drops/min** — Step 1: 120 mL/hour = 120 mL / 60 min

Step 2: 120 / 60 = 2 mL/min

Step 3: 2 mL × 20 drops = 40 drops/min

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**Q1 · Page 209**

**3.33 mL/min** — 5 h = 300 min

1000 / 300 = 3.33 mL/min

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**Q3 · Page 209**

**360 mL** — 3 h = 180 min

2 mL/min × 180 min = 360 mL

## 8.3 Dose Rates (mcg/kg/min) & mL/hour

**Slide 7**

**a. ≈ 5.1 mL/hour** — 2 × 68 = 136 mcg/min
136 × 60 = 8160 mcg/hour = 8.16 mg/hour
800 mg → 500 mL
8.16 mg → x
x = 8.16 × 500 / 800 = 5.1 mL/hour (slide: ≈ 5 mL/hour)

**b. ≈ 1.7 drops/min** — 5.1 mL/hour × 20 = 102 drops/hour
102 / 60 = 1.7 drops/min (slide: 100 / 60 = 1.7)

**Slide 8**

**≈ 4.2 mcg/kg/min** — 250 mg → 50 mL
x → 3.5 mL → x = 17.5 mg/hour
17.5 / 60 = 0.2917 mg/min = 291.7 mcg/min
291.7 / 70 = 4.17 ≈ 4.2 mcg/kg/min

## 8.4 Volume, Drip Rate & Duration

**Slide 9**

**500 min = 8 h 20 min** — 1000 mL × 20 = 20 000 drops
20 000 / 40 = 500 min
500 / 60 = 8.33 hours = 8 h 20 min

**Example 11.2 · Page 196**

**a. 100 mL** — 5 mg → 1 mL
500 mg → y → y = 100 mL

**b. 60 drops/min** — 3 mL/min × 20 = 60 drops/min

**c. 33.3 min** — 100 mL / 3 mL per min = 33.3 min

**Q5 · Page 209**

**a. 3 vials** — One vial = 25 × 4 = 100 mg
300 / 100 = 3 vials

**b. 5 mL/hour** — Concentration = 300 mg / 150 mL = 2 mg/mL
10 mg/hour ÷ 2 mg/mL = 5 mL/hour

**c. 30 hours** — 150 mL / 5 mL per hour = 30 h (or 300 mg / 10 mg per hour)

## 8.5 Dose, Volume & Maximum Rate

**Slide 10**

1. **29 200 mcg = 29.2 mg** — 400 mcg × 73 kg = 29 200 mcg
2. **≈ 194.7 mL** — Maximum concentration 150 mcg/mL:
   29 200 / 150 = 194.7 mL
3. **≈ 0.93 mL/min** — 3.5 h = 210 min
   194.7 / 210 = 0.927 mL/min
4. **800 mL** — 120 mg = 120 000 mcg
   120 000 / 150 = 800 mL

**Slide 11**

**a. 3.75 mL** — 75 / 20 = 3.75 mL

**b. 0.4 mL/min** — 8 mg/min ÷ 20 mg/mL = 0.4 mL/min

**c. ≈ 9.4 min** — 75 mg / 8 mg per min = 9.375 min (or 3.75 mL / 0.4 = 9.375 min)

**Example 11.4 · Page 198**

**a. 14 000 micrograms** — 250 × 56 = 14 000 micrograms

**b. 140 mL** — 14 000 / 100 = 140 mL

**c. 0.93 mL/min** — 2.5 h = 150 min

140 / 150 = 0.93 mL/min

**d. 500 mL** — 50 mg = 50 000 micrograms

50 000 / 100 = 500 mL (only 140 mL is given)

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**Example 11.6 · Page 201**

**a. 5 mL** — 50 / 10 = 5 mL

**b. 12.5 min · 0.4 mL/min** — 50 / 4 = 12.5 min

5 mL / 12.5 min = 0.4 mL/min

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**Q8 · Page 210**

**5 mL · 0.33 mL/min** — 75 / 15 = 5 mL

0.3 g/hour = 300 mg / 60 min = 5 mg/min

5 mg/min ÷ 15 mg/mL = 0.33 mL/min

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**Example 11.5 · Page 200**

**a. 900 mg · 18 mL** — 18 × 56 = 900 mg

900 / 50 = 18 mL

**b. Yes · 2.78 mL/min** — 18 mL fits into 100 mL (made up with 0.9% saline);

25 mg/min < 50 mg/min

900 mg → 100 mL

25 mg → x = 2.78 mL/min

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**Q6 · Page 209**

**a. 2 mL/min** — 1000 mg / 200 mL = 5 mg/mL

10 / 5 = 2 mL/min

**b. ≈ 1.67 mL/min** — 200 mL / 120 min = 1.67 mL/min

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**Q7 · Page 209**

**a. 1.36 mL** — Dose = 20 × 3.4 = 68 mg

68 / 50 = 1.36 mL

**b. 7.5 mL/min** — 3 × 3.4 = 10.2 mg/min

Concentration = 68 mg / 50 mL = 1.36 mg/mL

10.2 / 1.36 = 7.5 mL/min

**c. 0.833 mL/min** — 50 mL / 60 min = 0.833 mL/min

## 8.6 mg/hour → mL/min & drops/min

**Slide 12**

**0.33 mL/min · ≈ 7 drops/min** — 5 mg/hour ÷ 0.25 mg/mL = 20 mL/hour

20 / 60 = 0.33 mL/min

0.33 × 20 = 6.7 → 7 drops/min

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**Example 11.7 · Page 203**

**0.33 mL/min · 7 drops/min** — 3 mg/hour = 0.05 mg/min

0.05 / 0.15 = 0.33 mL/min

0.33 × 20 = 6.7 → 7 drops/min

Q9 · Page 210

**0.417 mL/min · 10.4 drops/min** — 0.05 mg/hour = 50 micrograms/hour
50 / 2 = 25 mL/hour
25 / 60 = 0.417 mL/min
0.417 × 25 = 10.4 drops/min (≈ 10)

## 8.7 Syringe Drivers (mm/hour)

Example 11.8 · Page 204

**3 mm/hour** — 5 mL injection + 5 mL NaCl → 50 mg in 10 mL (60 mm)
50 mg → 60 mm
2.5 mg → x = 3 mm
So 3 mm/hour

Example 11.9 · Page 205

**4 vials · 16 mL diluent** — 100 / 5 = 20 mg/hour → 2 mL must contain 20 mg
20 mL → 200 mg
200 / 50 = 4 vials (4 mL)
Diluent = 20 − 4 = 16 mL

Q10 · Page 210

**7.5 mm/hour** — 5 mL × 4 = 20 mg in 60 mm
2.5 mg → 2.5 × 60 / 20 = 7.5 mm
So 7.5 mm/hour

Q11 · Page 210

**9.375 micrograms** — 10 mL × 5 = 50 micrograms in 80 mm
3 h × 5 mm = 15 mm
15 × 50 / 80 = 9.375 micrograms